MindWell: Mental Health Awareness

A

Major Project

Submitted in partial fulfilment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

IN

Computer Science And Applications

by

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(April 2026)

CERTIFICATE

This is to certify that the work titled "MindWell: A Digital Platform for Mental Health Awareness and Self-Care" submitted by "Harshu Sharma", (MCAN1CA24009) in partial fulfilment for the award of the degree of MCA. (CSE), ITM University, Gwalior has been carried out under my supervision.
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Acknowledgement

Firstly, I thank Lord Almighty for making it possible for me to complete this work. The success and outcome of this project required a lot of guidance and assistance from many people, and I am incredibly privileged to have got it all along with the completion of my project. All that I have done is only due to such supervision and assistance and I would not forget to thank them.

I respect and thank Mr. Ashish Yadav for providing us an opportunity to do the project work and giving us all support and guidance which made us complete the project duly. He took keen interest on our project work and guided us all along and provided all the necessary information for developing a good system.

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At last but not the least we would like to thank our parents who trusted us and helped us through the errand.

Abstract

Mental health has become an increasingly important concern in today's fast-paced and stressful world, with many individuals experiencing anxiety, depression, and emotional challenges. However, access to affordable, integrated, and user-friendly mental health support systems remains limited. Existing platforms often focus on isolated features and lack a holistic approach to mental wellness.

This project, titled "MindWell: A Comprehensive Mental Health Wellness Web Application," aims to address these challenges by providing a unified digital platform for mental health support. The application integrates multiple features, including a PHQ-9 based self-assessment tool for evaluating depression severity, a journaling system for emotional expression, guided meditation resources, audio-based relaxation content, and access to community support services.

The system is developed using modern web technologies such as React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for data storage. Redux Toolkit is used for efficient state management, and Chart.js is integrated to provide data visualization for tracking mental health trends.

A key aspect of the project is the implementation of a calm and emotionally supportive user interface, designed using principles of Human-Computer Interaction (HCI). This ensures that users feel safe, comfortable, and encouraged while interacting with the platform.

The proposed system demonstrates how technology can be effectively utilized to create an accessible, user-centric, and comprehensive mental health solution that promotes awareness, self-care, and overall well-being.

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Chapter 1

Introduction:

1.1 Background

Mental health has become a critical aspect of overall well-being in modern society. With increasing academic pressure, professional demands, and lifestyle challenges, individuals are more prone to experiencing stress, anxiety, and depression. According to global health studies, a significant portion of the population suffers from mental health issues but does not seek help due to lack of awareness, stigma, or limited access to professional resources.

With the rapid growth of technology, web-based platforms have emerged as effective tools to bridge the gap between individuals and mental health support systems. Digital solutions offer accessibility, convenience, and privacy, making them suitable for users who may hesitate to seek traditional forms of help.

In this context, MindWell is developed as a comprehensive mental health wellness web application. It integrates multiple features such as self-assessment tools, journaling, meditation, audio therapy, and community resources to provide a holistic approach to mental well-being.

1.2 Problem Statement

Despite the growing need for mental health support, many individuals lack access to affordable, integrated, and user-friendly platforms. Existing solutions are often fragmented, focusing only on specific aspects such as meditation or therapy, and do not provide a complete mental wellness ecosystem. Additionally, many applications fail to consider emotionally supportive design, which is crucial for user engagement.

Therefore, there is a need for a comprehensive system that combines assessment, self-help tools, and resources in a single platform while ensuring accessibility and a calming user experience.

1.3 Objectives

The primary objective of this project is to develop a mental health wellness platform specific objectives are:

- To develop a unified web-based platform for mental health support.
- To implement a PHQ-9 based self-assessment tool for depression analysis.
- To provide a journaling feature for emotional expression and self-reflection.
- To offer guided meditation, audio stories, and wellness exercises.
- To integrate data visualization techniques for tracking mental health trends.
- To design a calm, user-friendly, and emotionally supportive interface.

1.4 Scope of the Project

The scope of the project is focused on developing a web-based application that provides essential tools and resources for mental wellness. The system is designed to be accessible to users through modern web browsers, ensuring ease of use without requiring specialized hardware or software.

The application covers key areas such as mental health assessment, journaling, relaxation techniques, and access to support resources. It allows users to monitor their mental health status, express their emotions, and engage with wellness content.

However, the system is not intended to replace professional medical diagnosis or therapy. Instead, it serves as a supportive tool that promotes awareness, self-care, and early identification of mental health issues. Future enhancements may expand its scope to include advanced features such as AI-based recommendations, real-time counselling, and mobile application support.

Chapter 2

Literature Survey

2.1 Introduction

The domain of mental health has evolved significantly with the integration of digital technologies, particularly in the areas of web applications, mobile platforms, and intelligent systems. Modern digital mental health solutions incorporate techniques such as machine learning (ML), artificial intelligence (AI), and data analytics to provide personalized and scalable mental health support.

Recent studies highlight the growing demand for accessible mental health tools due to increasing stress, anxiety, and depression levels worldwide. Digital platforms aim to bridge the gap between individuals and professional care by offering self-help tools, therapy assistance, and real-time monitoring systems.

2.2 Existing Mental Health Applications

Several applications have been developed to support mental well-being:

- Headspace and Calm provide guided meditation and mindfulness exercises to reduce stress and improve focus.
- 7 Cups offers emotional support through trained listeners and therapists, enabling peer-based counselling.

While these platforms are effective, they suffer from limitations such as:

- Subscription-based access
- Limited feature integration
- Dependency on human interaction

2.3 Mental Health Assessment Tools

Standardized tools like the PHQ-9 (Patient Health Questionnaire-9) are widely used for depression screening .

Studies by Kroenke et al. demonstrate that PHQ-9 is reliable for:

- Detecting depression severity
- Monitoring treatment progress

Digital platforms integrating such tools allow users to perform self-assessment, improving early detection and awareness.

2.4 Role of Journaling in Mental Health

Journaling is a proven therapeutic technique for emotional regulation. Research shows that expressive writing helps:

- Reduce anxiety and stress
- Improve emotional clarity
- Enhance self-awareness

However, existing applications provide journaling as an isolated feature rather than integrating it with analytics or feedback systems.

2.5 Machine Learning in Mental Health

Machine Learning has become a key component in mental health systems. It is used for:

- Depression prediction using user behaviour and text analysis
- Sentiment analysis of journal entries or social media data
- Early detection of mental disorders

Studies show that ML models such as:

- Support Vector Machines (SVM)
- Random Forest
- Deep Learning (LSTM, CNN)

can effectively classify mental health conditions based on textual and behavioural data

Datasets commonly used include:

- Social media datasets (Twitter, Reddit)
- Clinical survey datasets (PHQ-9 responses)
- User interaction logs

2.6 AI Chatbots in Mental Health

AI-powered chatbots have emerged as a scalable solution for mental health support. Examples include:

- Warbot – uses Cognitive Behavioural Therapy (CBT) techniques
- Wyse – provides AI-driven emotional support

These systems use:

- Natural Language Processing (NLP)
- Sentiment detection
- Conversational AI

Benefits:

- 24/7 availability
- Immediate response
- Cost-effective support

Limitations:

- Lack of deep emotional understanding
- Limited handling of complex mental conditions

2.7 Recommender Systems in Mental Health

Recommender systems enhance user experience by providing personalized suggestions such as:

- Meditation sessions
- Journaling prompts
- Coping strategies

These systems use:

- Collaborative filtering
- Content-based filtering
- Hybrid recommendation models

Research indicates that personalized recommendations improve user engagement and mental health outcomes.

2.8 Data Visualization in Mental Health Systems

Data visualization helps users track mental health trends through:

- Mood graphs
- Stress level charts
- Assessment score tracking

Technologies such as Chart.js and D3.js are widely used for visualization. However, many existing platforms lack:

- Interactive dashboards
- Long-term trend analysis
- Insightful interpretation of data

2.9 Comparative Analysis of Existing Systems

Study / Application	Features	Technologies Used	Limitations
Headspace	Meditation, mindfulness	Mobile App, Audio guidance	Paid, limited personalization
Calm	Relaxation,sleep stories	Mobile App	Subscription-based
7 Cups	Online counselling	Human support system	Delay, cost
Woebot	AI chatbot therapy	NLP, AI	Limited emotional depth
ML-based models	Depression prediction	SVM, LSTM	Requires large datasets
Recommender systems	Personalized suggestions	ML algorithms	Data dependency

Table 1

2.10 Limitations of Existing Systems

From the literature, the following gaps are identified:

- Lack of integrated platforms combining multiple mental health features
- Limited use of AI and ML for personalization
- High cost and subscription barriers
- Insufficient data visualization and insights
- Poor focus on user-centred emotional design
- Limited use of real-time adaptive systems

2.11 Summary

The literature review indicates that although numerous digital mental health solutions exist, they lack integration, affordability, and intelligent personalization. The proposed system MindWell aims to overcome these limitations by:

- Integrating assessment, journaling, chatbot, and visualization
- Utilizing AI and ML techniques for personalization
- Providing a user-friendly and emotionally supportive interface
- Ensuring accessibility and affordability

Chapter 3

System Analysis

3.1 Introduction

System analysis is an essential phase in software development that involves understanding the requirements, identifying problems in existing systems, and proposing an improved solution. This chapter presents an analysis of the existing systems, their limitations, and the proposed system designed to overcome these challenges. It also outlines the functional and non-functional requirements of the system.

3.2 Existing System

The existing mental health support systems include various web and mobile applications that provide services such as meditation guidance, therapy sessions, and emotional support. Platforms like meditation apps and online counselling services focus on specific aspects of mental wellness.

However, these systems have several limitations:

- Most applications provide limited functionality, focusing only on meditation or therapy.
- Many platforms require paid subscriptions, restricting access for users.
- Lack of integration between features such as assessment, journaling, and support resources.
- Interfaces are often complex or not designed for emotional comfort.
- Absence of data visualization tools to track user progress.

As a result, users need to rely on multiple platforms, which reduces convenience and engagement.

3.3 Proposed System

The proposed system, MindWell, is a comprehensive mental health wellness web application designed to address the limitations of existing systems.

It integrates multiple features into a single platform, including:

- PHQ-9 based self-assessment for evaluating depression severity.
- Journaling feature for emotional expression and self-reflection.
- Meditation and audio stories for relaxation and stress relief.
- Wellness exercises to promote physical and mental health.
- Community support resources for external help.
- Data visualization (charts) to track mental health trends.

The system is designed with a calm and user-friendly interface, ensuring that users feel safe and comfortable while using the application. It follows a modular architecture, making it scalable and easy to maintain.

3.4 Functional Requirements

Functional requirements define the core features and operations of the system.

- User registration and login functionality.
- Secure authentication and session management.
- PHQ-9 assessment submission and result calculation.
- Storage and retrieval of user data.
- Journal entry creation, viewing, and deletion.
- Access to meditation videos and audio stories.
- Display of mental health data using charts.
- Navigation between different modules of the application.

3.5 Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system.

- Usability: The system should be simple, intuitive, and easy to use.
- Performance: The application should load quickly and respond efficiently.
- Scalability: The system should handle an increasing number of users.
- Security: User data should be protected using secure authentication mechanisms.
- Reliability: The system should function consistently without failures.
- Accessibility: The platform should be accessible across different devices and browsers.

3.6 System Feasibility

3.6.1 Technical Feasibility

The system is technically feasible as it uses widely adopted technologies such as React.js, Node.js, and MongoDB, which are well-supported and efficient.

3.6.2 Economic Feasibility

The project is cost-effective as it uses open-source tools and frameworks, reducing development and maintenance costs.

3.6.3 Operational Feasibility

The system is user-friendly and can be easily operated by individuals without technical knowledge, making it highly practical for real-world use.

Chapter 4

System Design

4.1 Introduction

System design is a crucial phase in the development process that focuses on defining the architecture, components, and data flow of the system. It provides a blueprint for implementing the application and ensures that all system requirements are properly addressed. This chapter describes the overall design of the MindWell application, including architecture, database design, and system workflow.

4.2 System Architecture

The MindWell application follows a three-tier architecture, which includes:

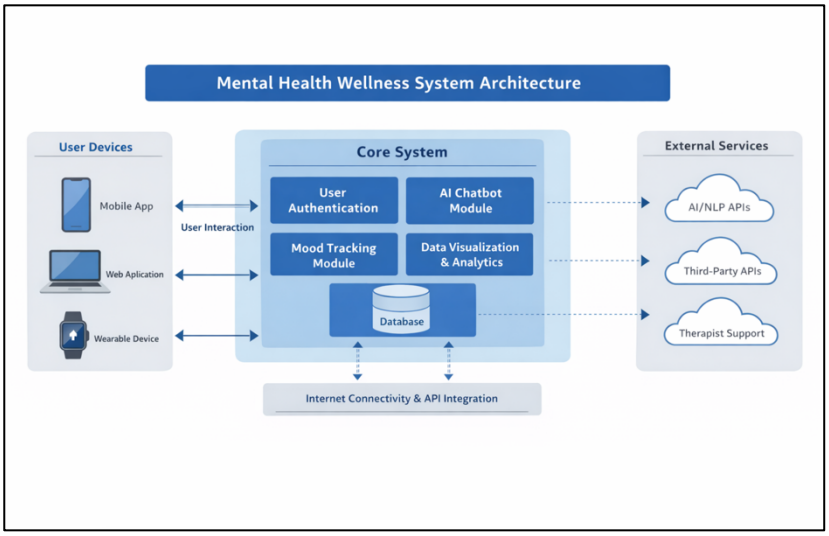
1. Presentation Layer (Frontend)
2. Application Layer (Backend)
3. Data Layer (Database)

Frontend (Presentation Layer)

- Displaying the user interface
- Handling user interactions
- Navigating between different pages
- Sending requests to the backend

Backend (Application Layer)

- API handling and routing
- Business logic processing
- Authentication and authorization
- Communication with the database



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Database (Data Layer)

The database used is MongoDB, which stores:

- User information (name, email, password)
- PHQ-9 assessment responses and scores
- Journal entries

This layered architecture ensures scalability, maintainability, and separation of concerns.

4.3 System Modules

The system is divided into several functional modules:

4.3.1 Authentication Module

- User registration and login
- Secure authentication using tokens
- Session management

4.3.2 Assessment Module

- PHQ-9 questionnaire
- Score calculation
- Severity classification
- Data storage and retrieval

4.3.3 Journal Module

- Add, view, and delete journal entries
- Store user thoughts and emotions

4.3.4 Wellness Module

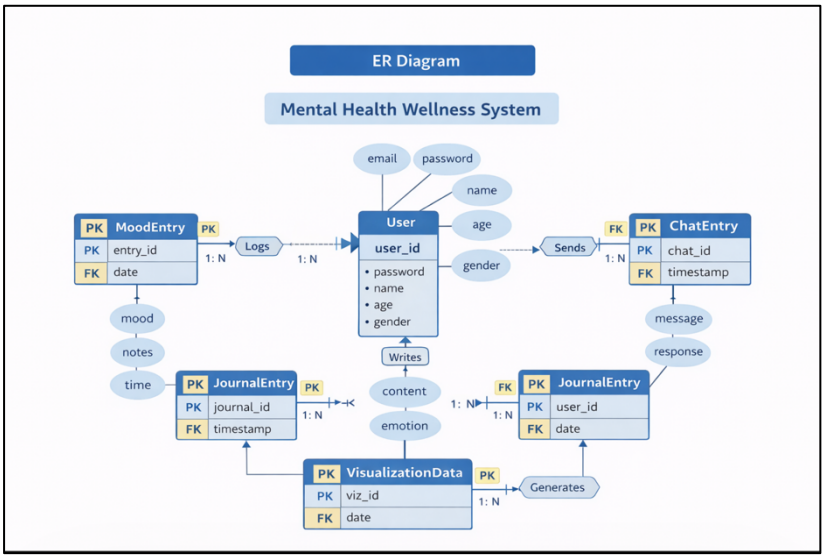
- Meditation videos
- Audio stories
- Exercise recommendations

4.3.5 Visualization Module

- Display of user mental health data
- Pie charts using Chart.js

4.3.6 Community Module

- External support resources and links



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4.4 Database Design

The system uses a NoSQL database (MongoDB) with the following collections:

User Collection

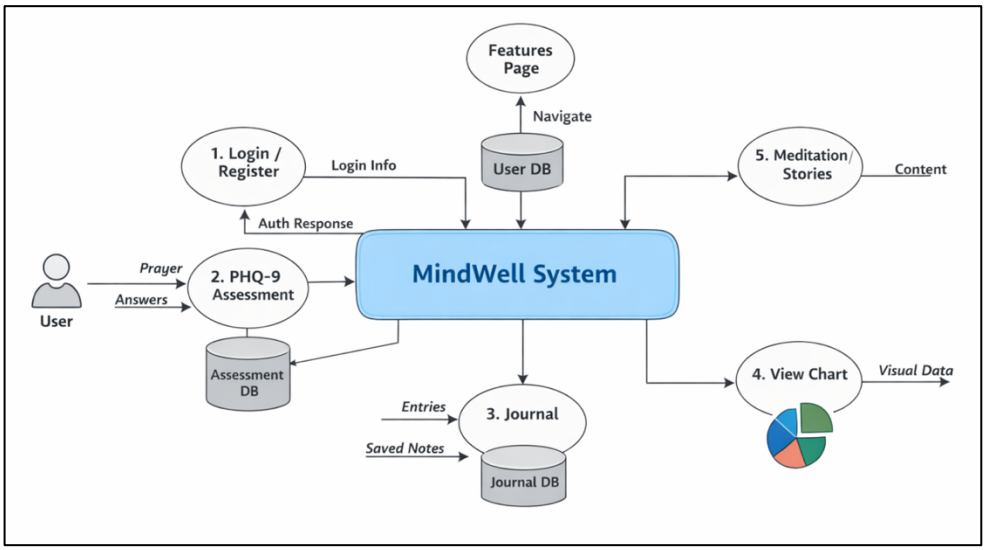
- User ID
- Name
- Email
- Password

PHQ9 Collection

- User ID
- Date
- Score

Journal Collection

- User ID
- Date
- Content



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This design ensures efficient data storage and retrieval.

4.5 Data Flow

The system follows a simple workflow:

1. User registers/logs in.
2. User accesses different modules (assessment, journal, etc.).
3. Data is sent to the backend via APIs.
4. Backend processes the request and stores data in MongoDB.
5. Data is retrieved and displayed to the user.
6. Visualization tools display results in chart format.

4.6 Design Considerations

- User-centred Design: Focus on simplicity and emotional comfort.
- Responsive Design: Works on different screen sizes.
- Modular Design: Each feature is implemented as a separate component.
- Security: User data is protected through authentication.

Chapter 5

Implementation

5.1 Introduction

The implementation phase involves the actual development of the system based on the design specifications. In this project, the MindWell application is developed using modern web technologies, ensuring a scalable, responsive, and user-friendly platform. This chapter describes how different components of the system are implemented, including frontend, backend, database integration, and key functionalities.

5.2 Development Environment

The system is developed using the following tools and technologies:

- Frontend: React.js, Tailwind CSS
- Backend: Node.js, Express.js
- Database: MongoDB
- State Management: Redux Toolkit
- Data Visualization: Chart.js
- Version Control: Git and GitHub
- Development Tools: Visual Studio Code

These technologies provide flexibility, performance, and ease of development.

5.3 Frontend Implementation

The frontend of the application is developed using React.js, which allows the creation of reusable components and dynamic user interfaces.

Key frontend features include:

- Responsive UI: Designed using Tailwind CSS for adaptability across devices.
- Navigation: Implemented using React Router for smooth page transitions.
- Component-Based Design: Each feature (Home, Features, Journal, Meditation, etc.) is built as a separate component.
- Animations: Framer Motion is used to enhance user interaction with smooth animations.
- User Interface Design: Soft colour schemes and minimal layout are used to create a calm and emotionally supportive experience.

5.4 Backend Implementation

The backend is developed using Node.js and Express.js, which handle server-side logic and API requests.

Key backend functionalities include:

- RESTful APIs: Used for communication between frontend and backend.
- Routing: Separate routes are defined for users and PHQ-9 assessments.
- Authentication: Secure login and registration system using tokens.
- Middleware: Used for handling requests, responses, and error management.
- Database Connectivity: Mongoose is used to interact with MongoDB.

5.5 Database Implementation

The database is implemented using MongoDB, which stores user-related data efficiently.

- User Data: Includes name, email, and password.
- Assessment Data: Stores PHQ-9 responses, scores, and dates.
- Journal Entries: Stores user-written content with timestamps.

The use of MongoDB ensures flexibility and scalability in handling structured and semi-structured data.

5.6 State Management

The application uses Redux Toolkit for managing global state.

- Handles user authentication state (login/logout)
- Stores user information across components
- Ensures consistent and predictable state updates

5.7 Data Visualization Implementation

Chart.js is integrated to display mental health insights.

- Pie charts are used to represent depression severity levels
- Data is fetched from the backend and processed for visualization
- Helps users understand their mental health trends easily

5.8 Key Functional Features Implementation

PHQ-9 Assessment

- Users answer a set of questions
- Score is calculated automatically
- Severity level is displayed

Journal System

- Users can write and save entries
- Entries are stored in local storage/database
- Users can delete entries

Meditation & Wellness

- Displays meditation videos
- Provides audio stories and exercises

Authentication System

- User registration and login
- Secure access to features

5.9 Testing and Debugging

The system is tested for:

- Functionality of each module
- API responses and database operations
- User interface responsiveness
- Error handling and validation

Debugging tools and console logs are used to identify and fix issues during development.

Chapter 6

Results and Discussion

6.1 Introduction

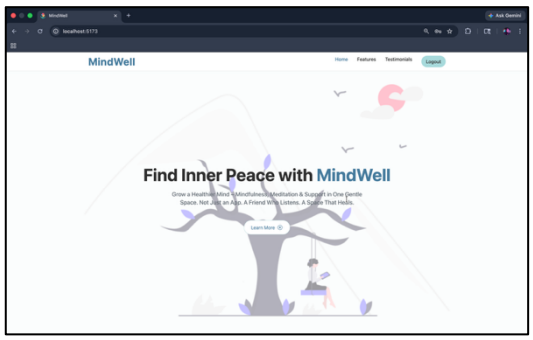
This chapter presents the results obtained from the implementation of the Mental Health Wellness System and discusses its performance, functionality, and effectiveness. The system was developed to provide users with mental health support through features such as mood tracking, AI chatbot interaction, journaling, and data visualization. The results demonstrate how the system performs in real-world scenarios and how effectively it meets the objectives defined in earlier chapters.

6.2 System Outputs

The developed system was successfully tested with different user inputs, and the following outputs were observed:

- User Authentication:
  Users were able to register and log in securely using authentication mechanisms.
- Mood Tracking:
  Users could input their daily mood, and the system stored and analysed this data effectively.
- AI Chatbot Interaction:
  The chatbot provided responses based on user queries related to mental health, offering basic guidance and support.
- Journaling Feature:
  Users were able to write and save personal thoughts, which helped in emotional expression and tracking mental patterns.
- Data Visualization:
  Graphs and charts were generated to show mood trends over time, helping users understand their mental health patterns.

6.3 Screen Shots



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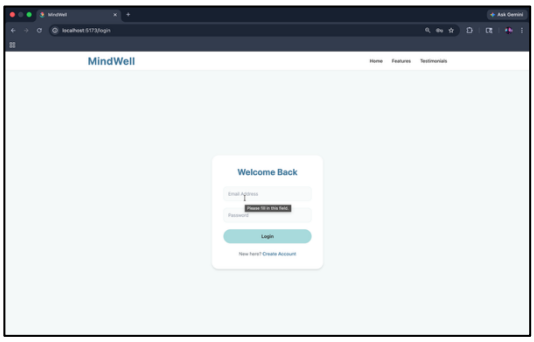

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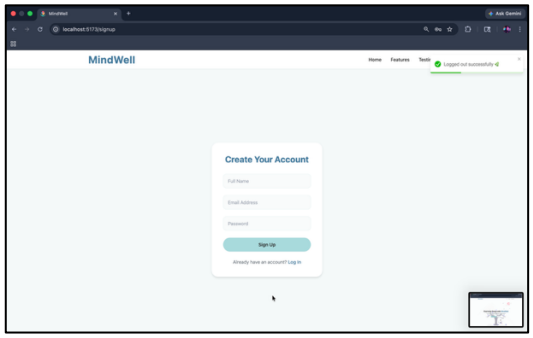

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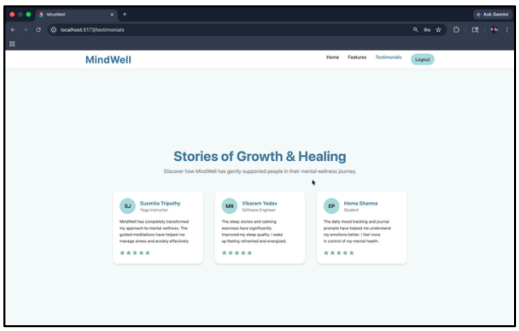

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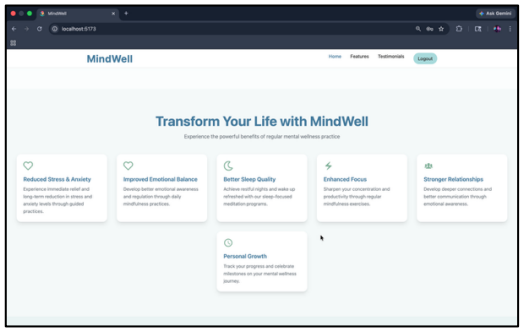

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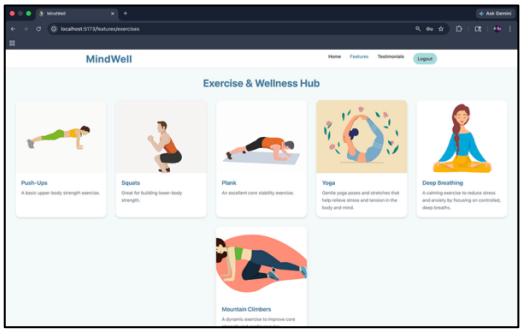

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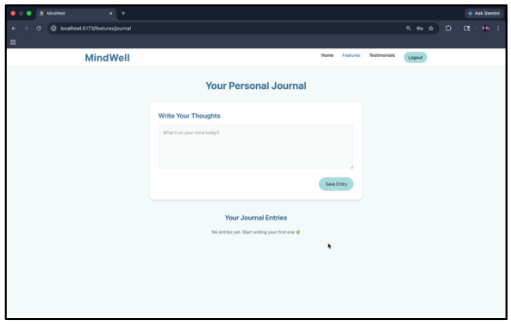

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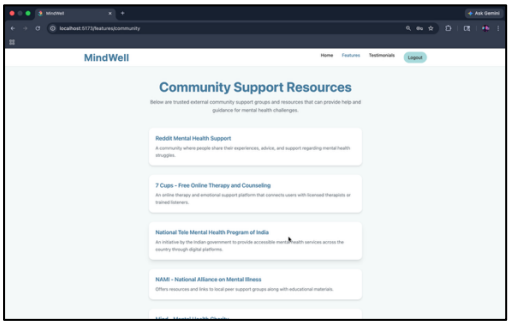

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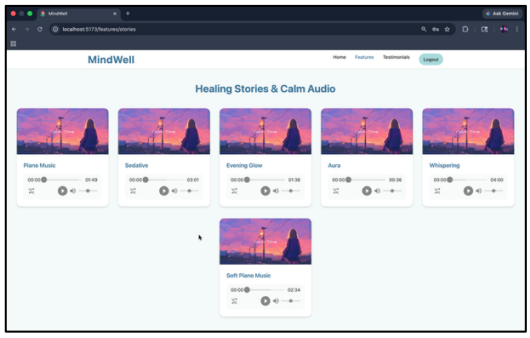

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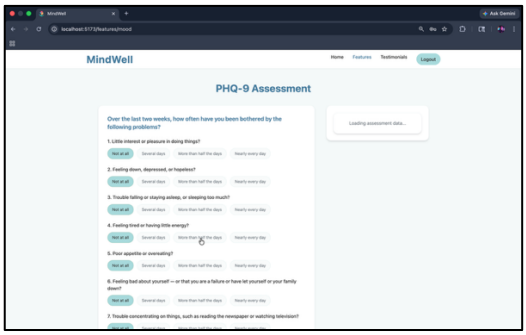

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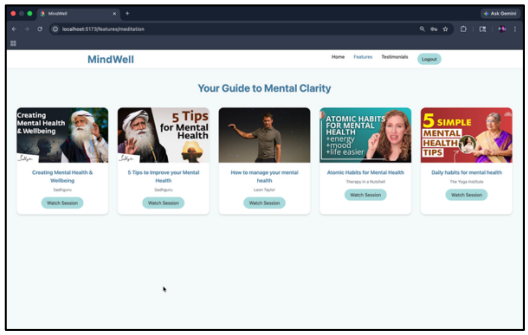

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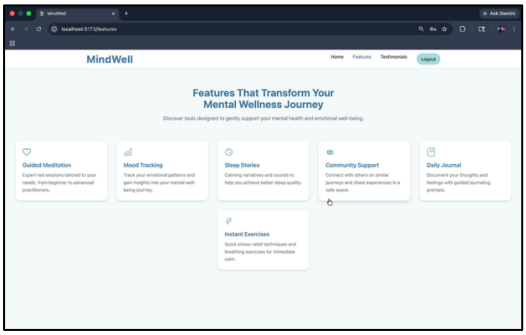

figure : 15

6.4 Performance Analysis

The system showed good performance in terms of:

- Response Time:
  The application responded quickly to user inputs, especially in chatbot interaction and data retrieval.
- Usability:
  The user interface was simple and user-friendly, making it easy for users to navigate through features.
- Scalability:
  The system can be extended to support more users and additional features in the future.
- Accuracy:
  The chatbot and recommendation system provided relevant responses based on available data and logic.

6.5 Functional Evaluation

The system was evaluated based on its core functionalities:

Feature	Expected Outcome	Actual Result
User Login/Register	Secure access	Successfully implemented
Mood Tracking	Store and analyse mood	Accurate tracking observed
Chatbot	Provide mental support	Basic support provided
Journaling	Store user thoughts	Successfully working
Visualization	Show trends	Graphs generated correctly

Table 2

All the major functionalities worked as expected without major errors.

6.6 Limitations

Despite successful implementation, some limitations were observed:

- The AI chatbot provides basic responses and is not as advanced as professional AI systems.
- The system depends on user input accuracy for mood tracking.
- Limited dataset for intelligent recommendations.
- No real-time integration with healthcare professionals.

6.7 Comparison with Existing Systems

The results indicate that the system successfully fulfils its primary objective of providing a basic mental health support platform. It demonstrates how modern technologies such as web development, data visualization, and AI-based interaction can be combined to create a useful application.

The project highlights the importance of digital mental health solutions in today's fast-paced world. While the system is not a replacement for professional therapy, it acts as a supportive tool for users to monitor and improve their mental well-being.

Chapter 7
Conclusion and Future Scope

7.1 Conclusion

The Mental Health Wellness System was developed to provide a simple, accessible, and effective platform for supporting users in monitoring and improving their mental well-being. The project successfully integrates multiple features such as mood tracking, journaling, AI-based chatbot interaction, and data visualization into a single system.

Through the implementation of this system, important concepts of web development, database management, and basic artificial intelligence were applied in a real-world context. The system allows users to record their emotions, analyze their mental health patterns, and receive basic guidance, thereby promoting self-awareness and emotional well-being.

The project also demonstrates how digital technologies can play a significant role in addressing mental health challenges in modern society. By providing an easy-to-use interface and essential features, the system ensures accessibility for a wide range of users.

Overall, the objectives of the project were successfully achieved. The system is functional, user-friendly, and capable of delivering meaningful insights into users' mental health. It serves as a strong foundation for further development and enhancement in the field of digital mental health solutions.

7.2 Future Scope

Although the current system performs effectively, there are several opportunities for future improvements and enhancements:

- Advanced AI Chatbot Integration:
  Implement more intelligent and context-aware chatbot models using advanced machine learning or natural language processing techniques for better interaction.
- Mobile Application Development:
  Develop a mobile app version (Android/iOS) to improve accessibility and user engagement.
- Integration with Wearable Devices:
  Connect with smart devices (like fitness bands or smartwatches) to collect real-time health data such as sleep patterns and activity levels.
- Real-time Therapist Support:
  Provide options to connect users with professional mental health experts for counselling and therapy.
- Personalized Recommendations:
  Use machine learning algorithms to provide personalized suggestions based on user behaviour and mood history.
- Enhanced Data Security:
  Implement advanced security mechanisms such as encryption and secure authentication to protect sensitive user data.
- Multilingual Support:
  Add support for multiple languages to make the system accessible to a wider audience.
- Community Support Features:
  Introduce forums or peer support systems where users can share experiences and support each other.

9. REFERENCES

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